ELEC5208 Active Distribution Systems

Group Assignment

TASK III: Energy System Optimisation

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CONTENT

1. HEMS Optimisation Formulation 3
1.1 System Description 3
1.2 Tariff Structure 3
1.3 Decision Variables 3
1.4 Constraints 4
1.5 Objective Function 5
2. Single-Node 24-Hour Dispatch Analysis 5
3. Full-Year Network Simulation — Cases 2 and 3 9
3.1 Case Definitions 9
3.2 Annual Performance Metrics 9
3.3 Voltage Analysis 11
3.4 Voltage Unbalance Analysis 12
4. Dynamic Operating Envelopes — Case 4 15
4.1 Methodology: SOCP-OPF 15
4.2 DOE Results 16
4.3 Location-Dependent DOE Analysis 17
4.4 Full-Year OpenDSS Verification of Case 4 18
5. Comparative Summary Across All Cases 19
6. Critical Discussion 19
6.1 SOCP Relaxation Tightness 19
6.2 Representative Period Limitation 20
6.3 Uniform vs. Differentiated DOE 20
6.4 SOC Continuity Between Days 20
6.5 Export Constraint and Arbitrage 21
6.6 Phase Imbalance and DOE Design 21
7. Conclusion 21
: 22

1. HEMS Optimisation Formulation

1.1 System Description

The Home Energy Management System (HEMS) is modelled as a linear programme (LP) solved over a rolling one-day horizon (288 × 5-minute steps, $\Delta t = 1/12$ h). Each of the 10 HEMS nodes (L10, L13, L16, L17, L22, L23, L24, L25, L26, L27) is equipped with a rooftop PV system and a battery (5 kW power rating, 90% round-trip efficiency), with capacities of 10 kWh or 20 kWh depending on the scenario.

1.2 Tariff Structure

The network operates under a Time-of-Use (ToU) tariff:

Period	Hours	Import Price ($/kWh)
Off-peak	00:00–06:59, 22:00–23:59	0.15
Shoulder	07:00–13:59, 20:00–21:59	0.25
Peak	14:00–19:59	0.35
Feed-in Tariff (FiT)	All hours	0.08 (export)

Table 1 Time-of-Use import tariff and feed-in tariff

Under the **Solar Sharer Offer (SSO)**, the import price during 11:00–13:59 is set to \$0.00/kWh, providing a free-electricity window during peak solar generation hours.

1.3 Decision Variables

At each timestep $t \in 1, \ldots, T$:

Variable	Description	Unit
$p_{import}[t] \geq 0$	Grid import power	kW
$p_{export}[t] \geq 0$	Grid export power	kW
$p_{pv}[t] \geq 0$	PV power delivered to loads/battery	kW
$p_{curt}[t] \geq 0$	PV curtailment	kW
$p_{charge}[t] \geq 0$	Battery charging power	kW

Variable	Description	Unit
$p_{discharge}[t] \geq 0$	Battery discharging power	kW
$SOC[t] \in [0, E_b at]$	State of charge	kWh

Table 2 HEMS optimisation decision variables and units

1.4 Constraints

PV energy balance — all available PV must be accounted for:

$$p_{pv}[t] + p_{curt}[t] \geq 0 = PV_{avail}[t] \forall t$$

Nodal power balance — instantaneous supply equals demand:

$$p_{import}[t] + p_{pv}[t] + p_{discharge}[t] = demand[t] + p_{charge}[t] + p_{export}[t]$$

Battery power limits:

$$p_{charge}[t] \leq P_{bat} = 5kW \forall t$$

$$p_{discharge}[t] \leq P_{bat} = 5kW \forall t$$

SOC dynamics ($\eta = 0.90$):

$$SOC[t+1] = SOC[t] + \eta \cdot p_{charge}[t] \cdot \Delta t - p_{discharge}[t] / \eta \cdot \Delta t \forall t$$

SOC bounds:

$$0 \leq SOC[t] \leq E_bat \quad \forall t$$

Import cap (prevents unbounded grid charging during off-peak):

$$p_i mport[t] \leq demand[t] + P_b at \forall t$$

Export cap (battery-to-grid arbitrage disallowed; export is PV-driven only):

$$p_{export}[t] \leq p_{pv}[t] \forall t$$

Terminal SOC constraint (applied only to the final day of the year to ensure annual energy balance):

$SOC[T+1] = SOC[1]$ (final day only)

Note on the export constraint: The constraint $p_{export}[t] \leq p_{pv}[t]$ intentionally prevents the battery from discharging directly to the grid. This reflects the modelling assumption that the battery serves local demand shifting rather than grid-export arbitrage. It should be noted that this restriction reduces the potential revenue from peak-export arbitrage and may understate the system's economic value in a more flexible market design.

1.5 Objective Function

The LP minimises the total electricity cost over the planning horizon:

$$min\Sigma_t[\lambda_{import}[t]\cdot p_{import}[t] - \lambda_F iT\cdot p_{export}[t] + \varepsilon\cdot(p_{charge}[t] + p_{discharge}[t])]\cdot\Delta t$$

where ε = 0.0001 $/kWh is a small cycling penalty to discourage unnecessary battery throughput without affecting dispatch decisions materially.

This is a linear programming problem because the objective function is linear in the decision variables and all listed constraints are linear equalities or inequalities. No binary variables are used in the base formulation. Therefore, the problem can be solved efficiently using a standard LP solver.

2. Single-Node 24-Hour Dispatch Analysis

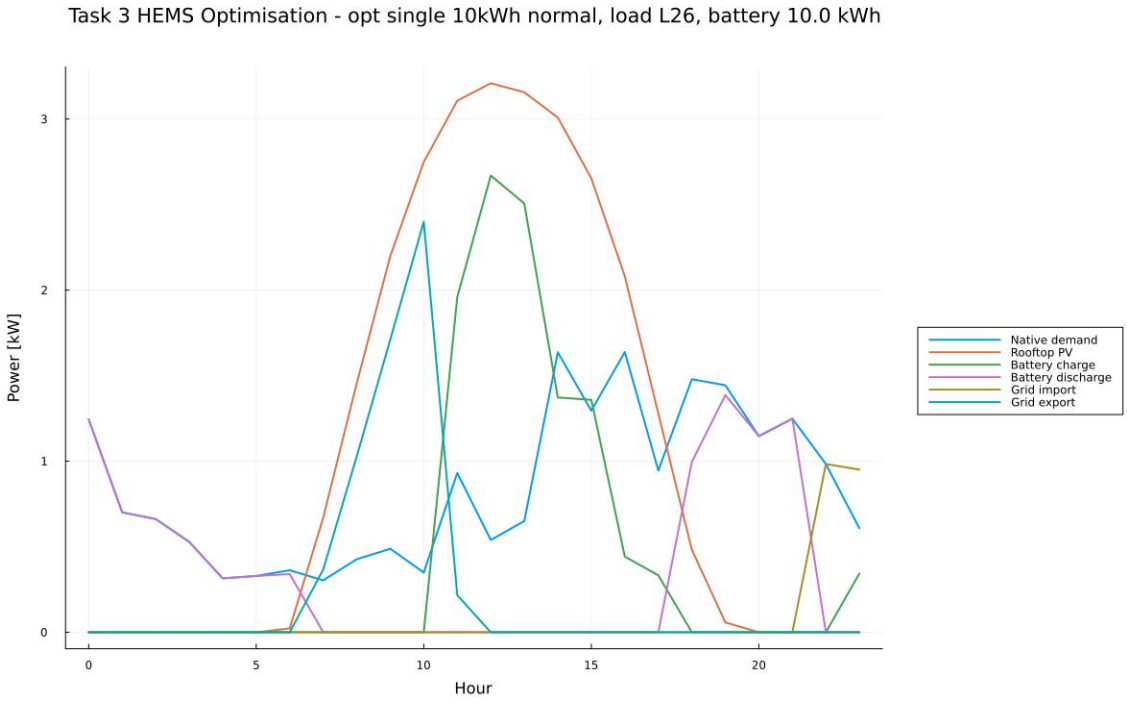


Figure 1 Single-node HEMS dispatch for 10 kWh battery under normal tariff

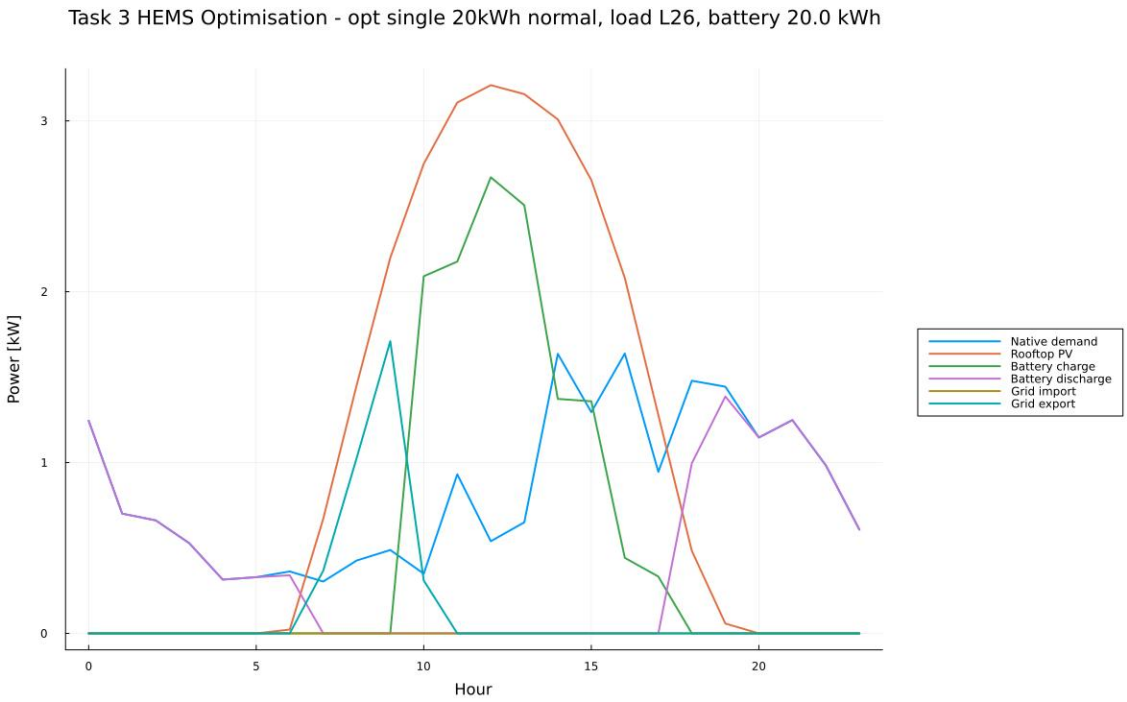


Figure 2 Single-node HEMS dispatch for 20 kWh battery under normal tariff

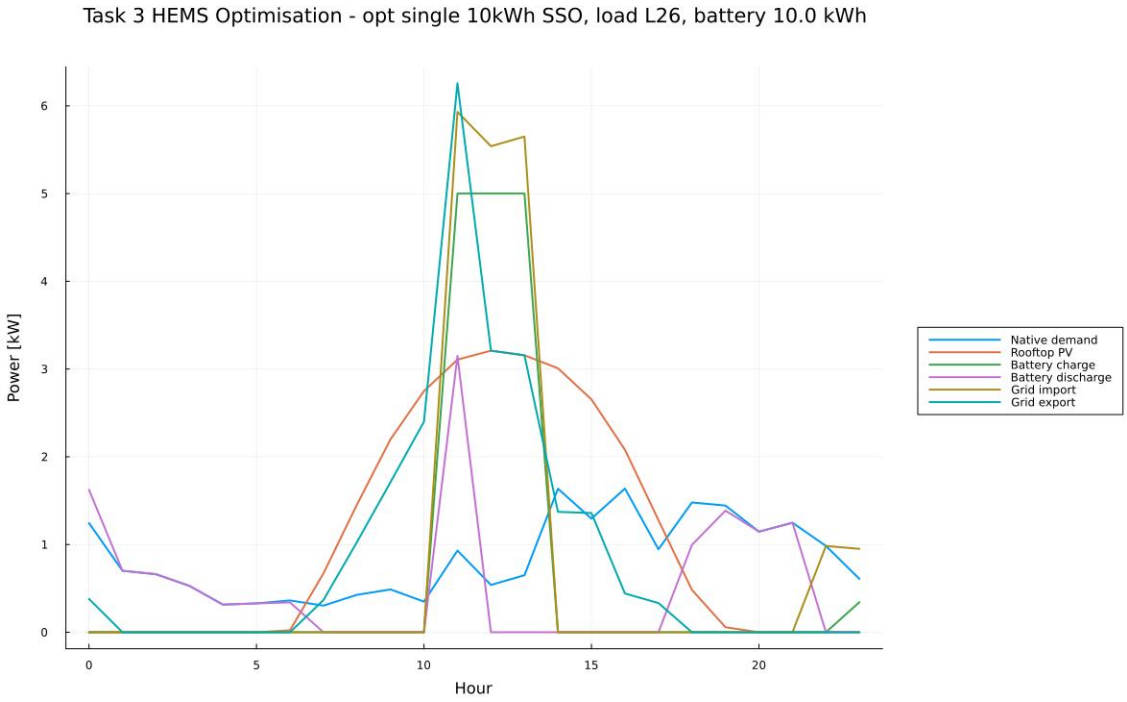


Figure 3 Single-node HEMS dispatch for 10 kWh battery under SSO tariff

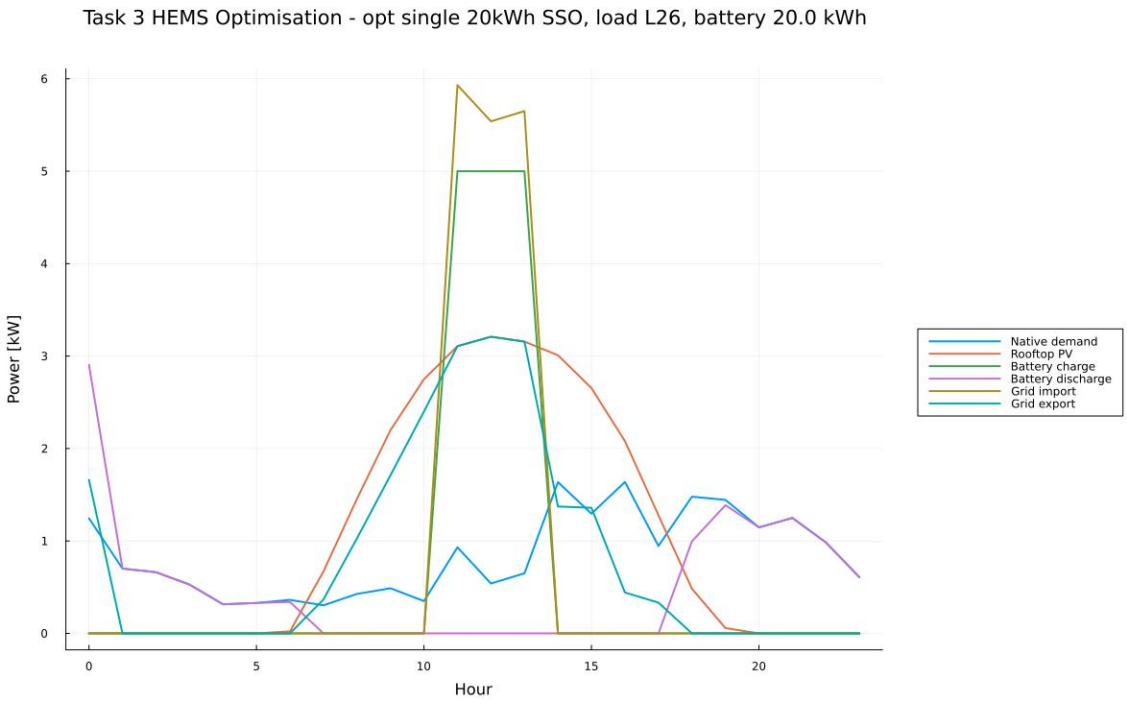


Figure 4 Single-node HEMS dispatch for 20 kWh battery under SSO tariff

Case	Net daily cost (AUD)	Grid import (kWh)	Grid export (kWh)	Battery charge (kWh)	Battery discharge (kWh)	PV curtailed (kWh)
10 kWh — Normal	**−0.17**	1.93	5.72	10.99	8.90	0.0
10 kWh — SSO	**−1.47**	19.06	22.02	15.34	12.43	0.0
20 kWh — Normal	**−0.27**	**0.00**	3.42	12.95	10.49	0.0
20 kWh — SSO	**−1.61**	17.12	20.15	15.00	12.15	0.0

Table 3 Comparison of 24-hour HEMS dispatch results under normal and SSO tariffs

The representative node L26 (bus B6a, phase A) is used. The LP dispatch exhibits three characteristic behaviours:

Morning charge phase (07:00–12:00): Surplus PV charges the battery preferentially over exporting, since the FiT (8 ¢/kWh) is well below the import price (25–35 ¢/kWh). Storing now avoids buying later.

Peak discharge phase (14:00–19:59): The battery discharges to meet the evening demand peak, avoiding the 35 ¢/kWh peak tariff. The optimiser times the discharge to coincide precisely with the price transition at 14:00.

Night idle: With no PV and low overnight demand, the battery holds its state unless off-peak charging is economical.

Effect of battery capacity (10 kWh vs 20 kWh): The larger battery can store more of the midday PV surplus, enabling greater peak-period self-sufficiency. This reduces grid import cost but increases capital expenditure — a trade-off examined in the full-year analysis.

Effect of SSO: The zero-price window (11:00–13:59) removes the incentive to export during those hours. Without SSO the LP exports PV surplus that cannot be stored; with SSO it is indifferent between export and storage during the free window, but avoids import cost at other times. The 20 kWh battery exploits SSO more effectively because it has capacity to absorb the entire midday surplus.

Critical observation: Under SSO, the zero import price can change the battery charging schedule if the battery has available capacity. In this implementation, grid import remains limited by the demand-plus-battery-power cap, and export is restricted to be PV-driven. Therefore, the model avoids unbounded free-electricity charging. However, because the formulation does not use binary import/export variables, simultaneous import and export is not explicitly forbidden. This is a modelling limitation and should be checked in the dispatch results. This means free electricity cannot always be fully exploited if the battery is already charged or the charger is at full rate — a practical limitation of the SSO design for prosumers with large PV systems.

3. Full-Year Network Simulation — Cases 2 and 3

3.1 Case Definitions

Case	Description
Case 2 — PV only	15 nodes with PV; no battery; net load = demand − PV
Case 2 SSO	Same as Case 2 (SSO has no effect without a battery)
Case 3 — 10 kWh opt	10 HEMS nodes with optimised 10 kWh battery
Case 3 — 20 kWh opt	10 HEMS nodes with optimised 20 kWh battery
Case 3 — 10 kWh SSO	10 HEMS nodes, 10 kWh battery, SSO tariff
Case 3 — 20 kWh SSO	10 HEMS nodes, 20 kWh battery, SSO tariff

Table 4 Case definitions for full-year OpenDSS simulations

3.2 Annual Performance Metrics

Metric	Case 2	Case 2 SSO	Case 3 10kWh	Case 3 20kWh	Case 3 10kWh SSO	Case 3 20kWh SSO
Min voltage (p.u.)	0.9137	0.9137	0.8986	0.8855	0.8834	0.8834
Max voltage (p.u.)	1.0705	1.0705	1.0705	1.0705	1.0671	1.0635
Voltage violations (steps)	0	0	1	2	2	2
Max PVUR (%)	6.4668	6.4668	6.3669	7.3578	7.4144	7.4144
Max VUF (%)	3.3877	3.3877	3.3878	3.7432	3.7322	3.7322
Reverse power flow hours	1586.00	1586.00	1138.58	1132.58	1078.67	1037.67
Annual feeder import (MWh)	152.936	152.936	147.603	147.909	147.309	147.489
Annual feeder export (MWh)	24.086	24.086	13.664	13.414	12.002	11.346

Table 5 Annual network performance comparison for Cases 2 and 3

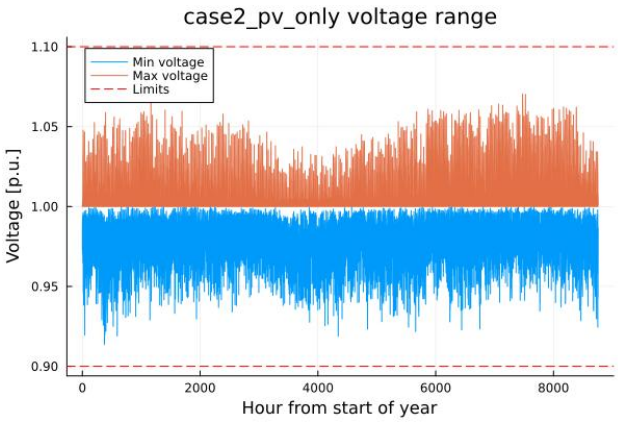


Figure 5(a) Annual voltage range for Case 2 — PV only

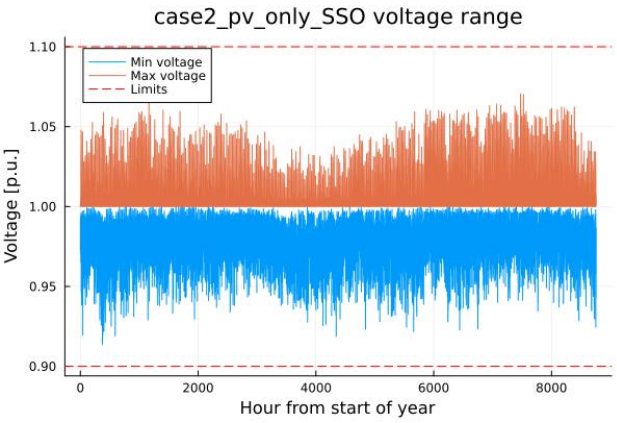


Figure 5(b) Annual voltage range for Case 2 — SSO tariff

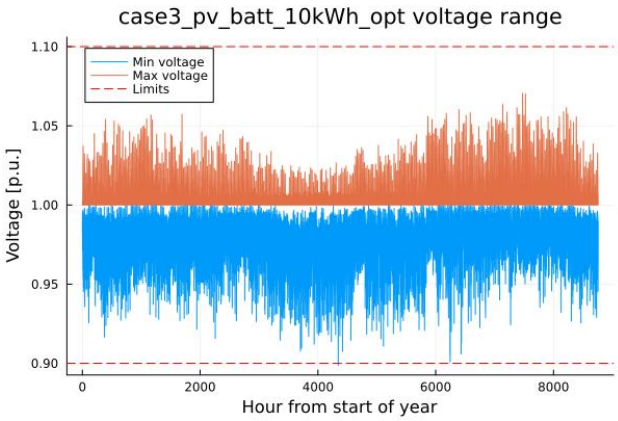


Figure 5(c) For Case 3 — 10 kWh optimised battery

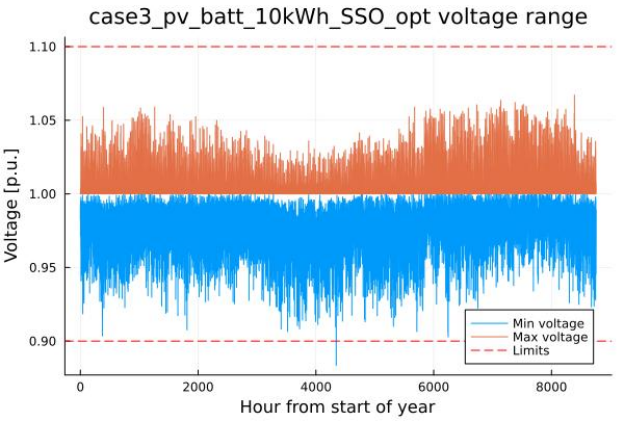


Figure 5(d) For Case 3 — 10 kWh battery with SSO tariff

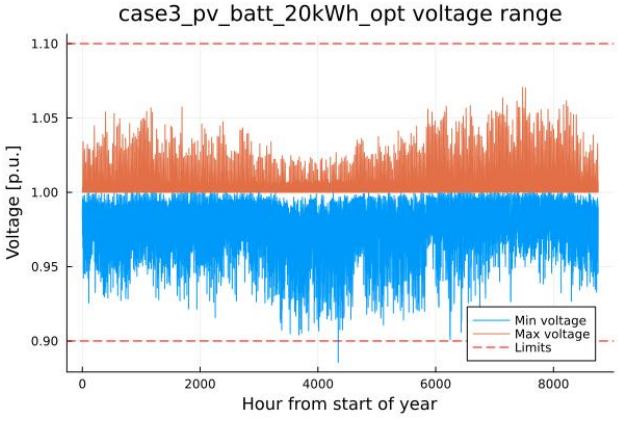


Figure 5(e) For Case 3 — 20 kWh optimised battery

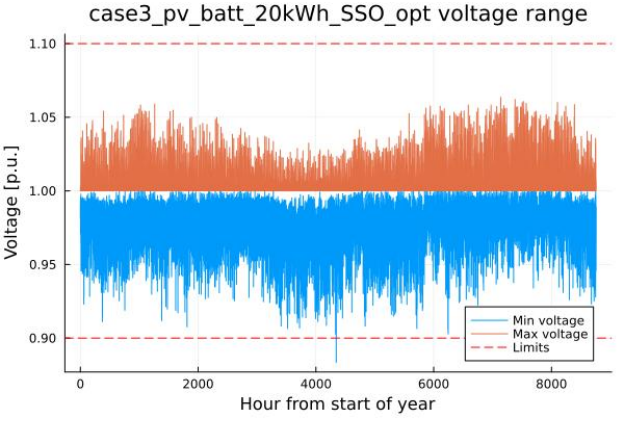


Figure 5(f) For Case 3 — 20 kWh battery with SSO tariff

3.3 Voltage Analysis

The annual voltage results show that all Case 2 time steps remained within the 0.9–1.1 p.u. voltage limits. In the PV-only case, the minimum voltage was 0.9137 p.u. and the maximum voltage was 1.0705 p.u. Although no voltage magnitude violations occurred, the maximum voltage was higher than the minimum-load periods without PV, indicating that rooftop PV export caused visible midday voltage rise.

Case 2 and Case 2 SSO produced identical voltage results. This is because the SSO tariff only changes the import price signal, but there is no battery or other controllable resource in Case 2. Therefore, the physical net-load profile remains unchanged, and the OpenDSS power-flow results are also unchanged.

In the Case 3 scenarios, the optimised batteries reduced annual feeder export and reverse power flow hours by storing part of the midday PV surplus. For example, annual feeder export decreased from 24.086 MWh in Case 2 to 13.664 MWh in Case 3 with the 10 kWh battery, and to 13.414 MWh with the 20 kWh battery. The SSO tariff further reduced export to 12.002 MWh for the 10 kWh battery and 11.346 MWh for the 20 kWh battery. This confirms that the HEMS shifts part of the PV surplus away from direct grid export.

However, reducing export did not eliminate all voltage issues. Several Case 3 scenarios introduced rare undervoltage events. The minimum voltage fell to 0.8986 p.u. in the 10 kWh battery case, and to 0.8855–0.8834 p.u. in the 20 kWh and SSO cases. Since the maximum voltages remained below 1.1 p.u. in all Case 3 scenarios, these violations were caused by low-voltage events rather than PV-driven overvoltage. This indicates that battery operation can reduce export-driven voltage rise, but it may also shift demand and charging behaviour in ways that worsen low-voltage conditions during some periods.

Overall, the voltage results show that HEMS optimisation improves export management and reduces reverse power flow, but it is not a complete voltage-regulation solution. Additional network-level coordination, such as DOE constraints or voltage-support control, is still required to manage both overvoltage and undervoltage risks.

Critical observation — Case 2 SSO identical to Case 2: Without a battery, the SSO zero-price window provides no additional flexibility to the prosumer. The net load is physically unchanged regardless of price signals. This is confirmed by the identical DSS LoadShapes for the two cases and is physically correct, but it highlights a fundamental limitation: SSO benefits are *storage-conditional*.

3.4 Voltage Unbalance Analysis

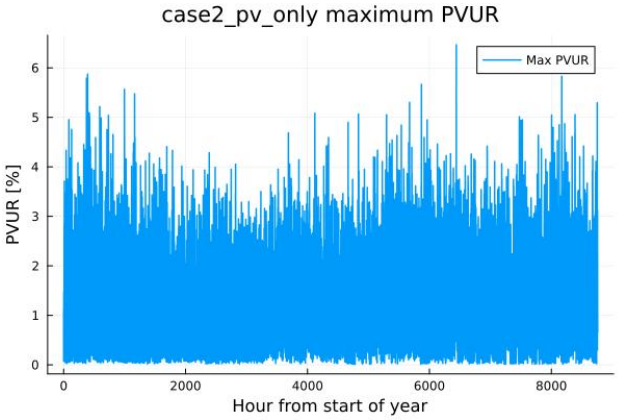


Figure 6(a) Maximum PVUR profile for Case 2 — PV only

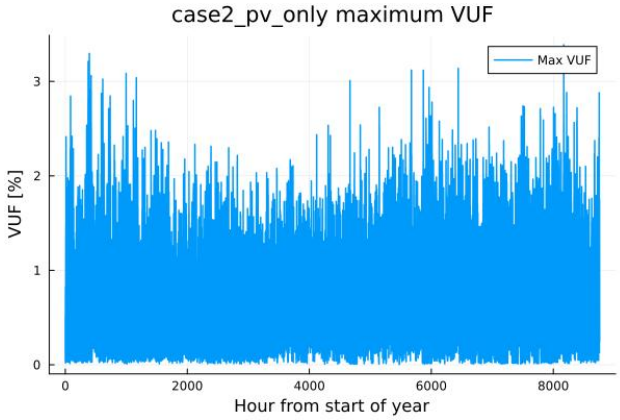


Figure 6(b) Maximum VUF profile for Case 2 — PV only

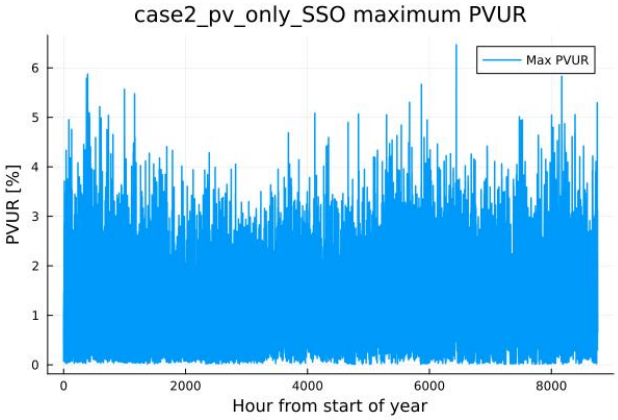


Figure 6(c) Maximum PVUR profile for Case 2 — SSO tariff

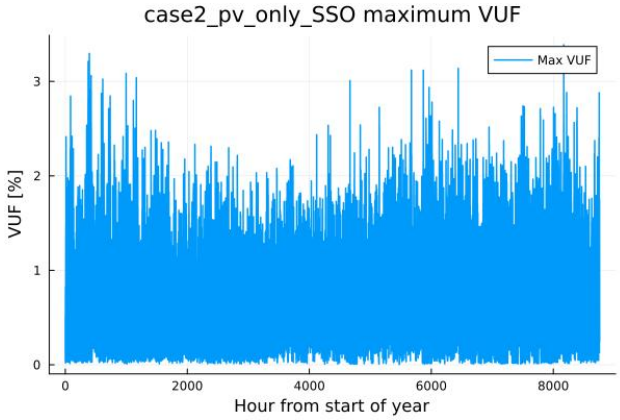


Figure 6(d) Maximum VUF profile for Case 2 — SSO tariff

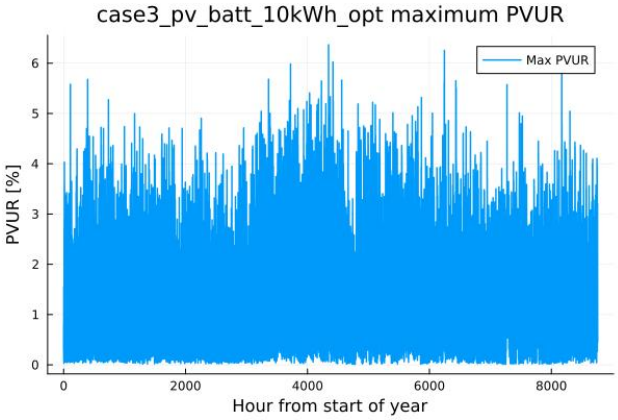


Figure 6(e) Maximum PVUR profile for Case 3 — 10 kWh

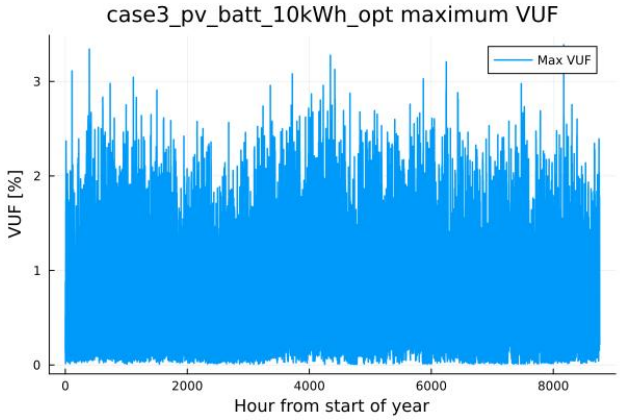


Figure 6(f) Maximum VUF profile for Case 3 — 10 kWh

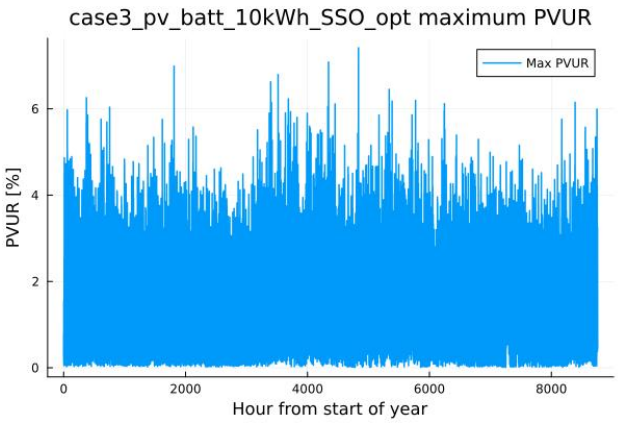


Figure 6(g) Maximum PVUR profile for Case 3: 10 kWh with SSO

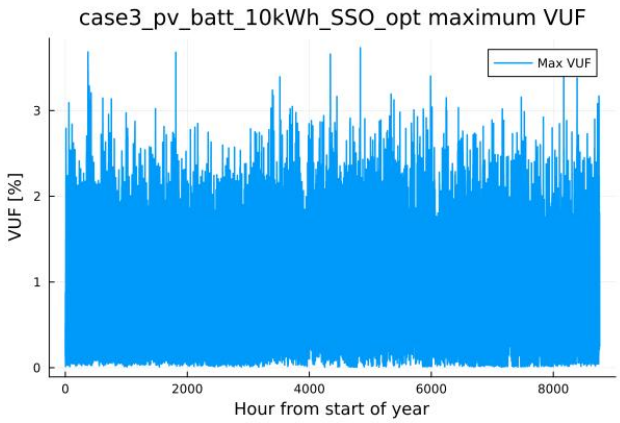


Figure 6(h) Maximum VUF profile for Case 3: 10 kWh with SSO

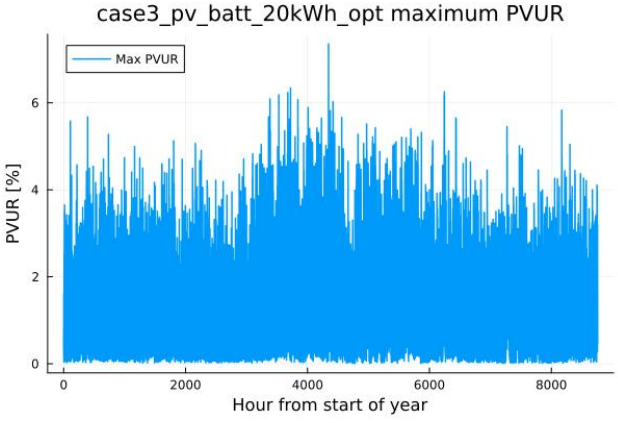


Figure 6(i) Maximum PVUR profile for Case 3 — 20 kWh

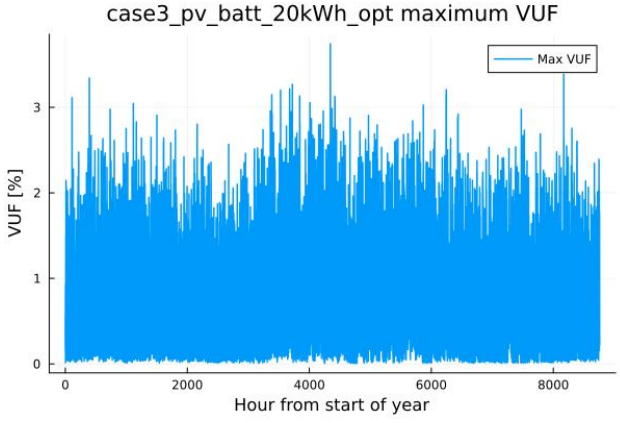


Figure 6(j) Maximum VUF profile for Case 3 — 20 kWh

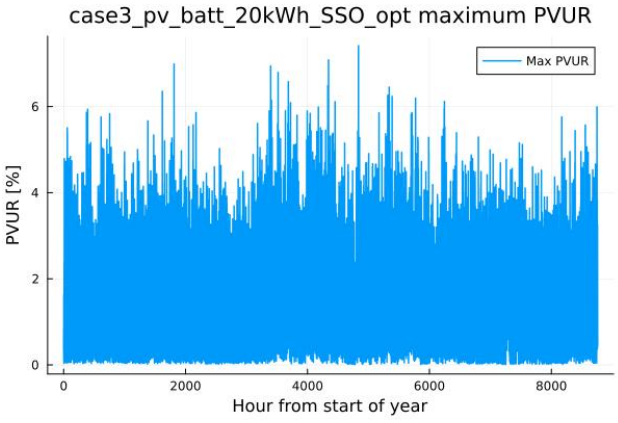


Figure 6(k) Maximum PVUR profile for Case 3: 20 kWh with SSO

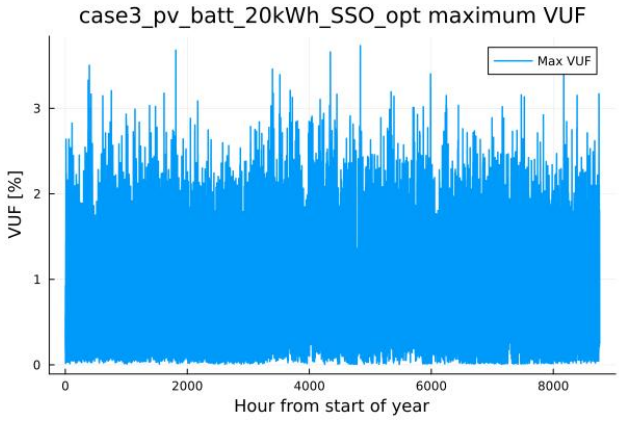


Figure 6(l) Maximum VUF profile for Case 3: 20 kWh with SSO

The annual PVUR and VUF results show that voltage unbalance remains an important issue in the LV feeder. In Case 2, the maximum PVUR and VUF were 6.4668% and 3.3877%,

respectively. Since Case 2 SSO has no battery or controllable resource, it produced identical PVUR and VUF values to Case 2.

The Case 3 results show that optimised battery operation does not necessarily reduce the worst-case voltage unbalance. Although the HEMS reduces annual feeder export and reverse power flow, the maximum unbalance indices remain similar to, or higher than, those in Case 2. For example, the 10 kWh battery case slightly reduced the maximum PVUR to 6.3669%, but the 20 kWh and SSO cases increased the maximum PVUR to around 7.36–7.41%. Similarly, the maximum VUF increased from 3.3877% in Case 2 to approximately 3.73–3.74% in several Case 3 scenarios.

This occurs because voltage unbalance is strongly affected by the phase allocation and location of single-phase PV and battery systems. Individual HEMS optimisation mainly minimises household energy cost and shifts import/export timing, but it does not explicitly coordinate the three phases. As a result, battery charging or discharging on one phase can reduce feeder export while still increasing phase-specific voltage differences at certain times.

Therefore, the results indicate that HEMS optimisation is effective for reducing reverse power flow and annual export, but it is not sufficient as a phase-balancing strategy. Additional network-level coordination, such as phase-aware DOE allocation, inverter reactive power control, or explicit unbalance constraints in the optimisation, would be required to systematically reduce PVUR and VUF.

4. Dynamic Operating Envelopes — Case 4

4.1 Methodology: SOCP-OPF

The DOE for each HEMS node is computed by solving a **Branch Flow Model (BFM) SOCP relaxation** for each hour of a representative 7-day period (168 hourly problems × 3 phases = 504 SOCP solves).

Because the DOE calculation uses a simplified per-phase SOCP model, the DOE values should be interpreted as approximate optimisation-based export envelopes. The full three-phase four-wire feasibility is checked later using OpenDSS verification.

Network model (single-phase, per-phase independent):

The LV feeder is modelled as a radial network with one slack bus (Tbus at the transformer secondary, V = 1.0 p.u.) and 9 non-slack buses. Line parameters are derived from the Neptune conductor (R = 0.224 Ω/km, X = 0.316 Ω/km from Carson's equations) and per-unit base S_1 = 500/3 = 166.7 kW, V_1 = 230 V, Z_1 = 0.317 Ω, I_1 = 724.6 A.

Decision variables (per bus i, per branch ij):

Variable	Description
$u_i = \|V_i\|^2 \in [0.81, 1.21] p.u.^2$	Squared voltage magnitude
$\ell_i j \in [0, 0.437] p.u.^2$	Squared current magnitude (limit = (479/724.6)² = 0.437)
$P_i j, Q_i j$	Active and reactive branch power flows (p.u.)
$e_k \geq 0$	Export allowance for HEMS node k (p.u.)

Table 6 Decision variables of the SOCP-OPF model for DOE calculation

SOCP relaxation (Rotated Second-Order Cone):

$$\ell_i j \cdot u_i \geq P_i j^2 + Q_i j^2 \forall (i, j)$$

Distflow voltage recursion:

$$u_j = u_i - 2(r_i j \cdot P_i j + x_i j \cdot Q_i j) + (r_i j^2 + x_i j^2) \cdot \ell_i j \forall (i, j)$$

Nodal power balance (equality):

$$P_i j - r_i j \cdot \ell_i j - \Sigma_{jk} P_j k = p_j^b g + p_j^H EMS \forall j$$

where $p_j^b g$ is the hourly-average background (non-HEMS) net load from Case 3 profiles, and $p_j^H EMS = d_j - pv_j - e_j$ for HEMS nodes.

Objective — maximise total HEMS export:

$$max \Sigma_k e_k$$

The 7-day DOE table is then applied annually by cycling the 168-hour pattern.

4.2 DOE Results

Load	Bus	Phase	DOE Min	DOE Mean	DOE Max	Interpretation
L10	B4	A	119.81	137.15	153.43	Near transformer; effectively unlimited for 5 kW PV
L22	B4a	B	66.09	74.07	86.79	Lateral branch, phase B; moderate limit
L24	B4a	C	80.29	85.17	93.48	Lateral branch, phase C; ample headroom
L16	B6	B	54.80	60.14	73.22	Far-end, but phase B less congested
L27	B6a	C	37.05	44.36	52.02	End of feeder, phase C
L25	B6a	B	0.44	3.98	9.45	End of feeder, phase B; tightly constrained
L13	B5	A	1.30	2.52	3.27	Phase A congestion; binding on 5 kW PV
L17	B6	A	1.14	2.13	2.78	Phase A congestion; binding on 5 kW PV
L23	B4a	A	0.00	1.90	3.41	Phase A congestion; severely restricted
L26	B6a	A	1.13	2.05	2.67	Phase A end-of-feeder; severely restricted

Table 7 SOCP DOE limits by HEMS node (kW)

DOE values above the 5 kW residential inverter rating should be interpreted as non-binding limits, not expected household export power.

The DOE values in Table 7 represent the network export headroom calculated by the SOCP-OPF, rather than the actual export power of each household. In the subsequent HEMS dispatch and OpenDSS verification, the actual export is still limited by the household PV and inverter capability. Therefore, DOE values much larger than the 5 kW residential inverter rating, such as the 137.15 kW mean DOE for L10, should be interpreted as non-binding export limits. In contrast, DOE values below or close to 5 kW, such as those for L13, L17, L23 and L26, are binding or potentially binding for residential-scale PV export.

Load	Import (kWh)	Export (kWh)	DOE-curtailed (kWh)	Curtailment ratio
L10	958.4	3,605.7	0.0	0%
L13	2,163.2	1,915.3	870.2	31%
L16	3,153.8	1,951.7	0.0	0%
L17	5,901.9	1,177.9	357.5	23%
L22	2,163.2	2,785.6	0.0	0%
L23	3,899.5	1,757.8	3,867.2	69%
L24	436.7	3,115.1	0.0	0%
L25	3,153.8	1,300.1	651.6	33%
L26	6,020.6	1,159.4	407.9	26%
L27	1,281.2	5,229.2	0.0	0%

Table 8 Annual energy impact of DOE on HEMS loads

Curtailment ratio = curtailed / (export + curtailed)

4.3 Location-Dependent DOE Analysis

The results reveal a clear **network-position and phase-congestion pattern**:

Distance from transformer: L10 at B4 (fourth node from the transformer) obtains a DOE mean of 137 kW — far in excess of its 5 kW PV capacity. In contrast, L13 at B5 and L17 at B6 on the same phase obtain only 2.5 kW. This is explained by the Distflow voltage recursion: export at B4 raises voltage progressively at all downstream buses. Since voltage at B6a must remain below 1.1 p.u., and B4 is already at the limit, downstream nodes have essentially no further voltage headroom.

Phase congestion: Phase A carries the most HEMS load (L10, L13, L17, L23, L26 — five of the ten nodes), leading to the lowest DOE limits on phase A. Phases B and C carry fewer HEMS nodes and therefore enjoy more current and voltage headroom. L24 on phase C at B4a obtains a DOE of 85 kW despite being at the same bus as L23 (phase A, 1.9 kW).

Thermal vs. voltage binding: For nodes close to the transformer (L10), the binding constraint is the **branch thermal limit** (479 A) on branch B3–B4. For far-end nodes on phase A

(L13, L17, L26), the binding constraint is the **upper voltage limit** (1.1 p.u.), since voltage accumulates along the feeder.

4.4 Full-Year OpenDSS Verification of Case 4

Metric	Case 4 DOE
Min voltage (p.u.)	0.8855
Max voltage (p.u.)	1.0490
Voltage violations (steps)	2
Max PVUR (%)	7.3578
Max VUF (%)	3.7432
Reverse power flow hours	986.25
Annual feeder import (MWh)	149.828
Annual feeder export (MWh)	9.313

Table 9 OpenDSS verification results after applying DOE export limits

Compared with Case 3 20 kWh, the Case 4 DOE scenario provides an additional network-level export constraint on top of the individual HEMS optimisation.

The OpenDSS verification shows that applying the DOE export limits reduced the maximum feeder voltage to 1.0490 p.u. and reduced the annual feeder export to 9.313 MWh. Reverse power flow hours also decreased to 986.25 h, which is lower than Case 2 and the Case 3 HEMS-only scenarios. This confirms that the DOE is effective in limiting PV-driven voltage rise and reverse power flow.

However, two voltage violation steps remained because the minimum voltage reached 0.8855 p.u. Since the maximum voltage was well below the 1.1 p.u. upper limit, these remaining violations were undervoltage events rather than PV-driven overvoltage events. Therefore, the DOE should be interpreted as an export-management mechanism, not a complete voltage-regulation solution for all operating conditions.

5. Comparative Summary Across All Cases

Metric	Case 2	Case 3 10kWh	Case 3 20kWh	Case 4 DOE
Min voltage (p.u.)	0.9137	0.8986	0.8855	0.8855
Max voltage (p.u.)	1.0705	1.0705	1.0705	1.0490
Voltage violations (steps)	0	1	2	2
Reverse power flow hours	1586.00	1138.58	1132.58	986.25
Max PVUR (%)	6.4668	6.3669	7.3578	7.3578
Max VUF (%)	3.3877	3.3878	3.7432	3.7432
Annual feeder export (MWh)	24.086	13.664	13.414	9.313

Table 10 Comparative network performance across PV-only, optimised HEMS and DOE cases

The progression from Case 2 to Case 3 and Case 4 represents increasing levels of coordination: passive PV, individual HEMS optimisation, and network-level DOE control: passive PV, individual HEMS optimisation, and network-level DOE control. The results show that greater coordination consistently reduces annual feeder export and reverse power flow. However, not all voltage-quality indicators improve monotonically. In particular, undervoltage events and unbalance metrics remain in Case 3 and Case 4, indicating that export management alone is not sufficient to fully solve all network constraints.

6. Critical Discussion

6.1 SOCP Relaxation Tightness

The BFM SOCP model used in this study is a convex relaxation of the exact non-convex branch-flow power flow equations. The relaxation replaces the equality relationship between squared current, squared voltage, and branch power flow with a second-order cone inequality. Therefore, the optimisation problem can be solved efficiently, but the resulting DOE values should not automatically be assumed to be exact AC-feasible limits.

For radial distribution networks, SOCP relaxations are often tight under suitable operating conditions. However, the LV feeder in this study has resistance comparable to reactance and contains unbalanced single-phase PV, battery, and load injections. In addition, the DOE calculation was solved using a simplified per-phase model rather than a fully coupled three-phase four-wire AC model. These assumptions mean that the SOCP-derived DOE limits should be interpreted as optimisation-based export envelopes rather than guaranteed exact physical limits.

For this reason, an OpenDSS verification step was required after applying the DOE profiles to the full-year simulation. The verification showed that the DOE reduced the maximum voltage to 1.0490 p.u. and reduced annual feeder export to 9.313 MWh. However, two rare undervoltage violation steps remained, with the minimum voltage reaching 0.8855 p.u. This indicates that the DOE successfully mitigated PV-driven overvoltage and reverse power flow, but it did not fully address low-voltage events caused by demand-side loading.

6.2 Representative Period Limitation

The SOCP DOE was computed over only 7 days (the first week of January — Australian summer). Seasonal variation in PV output and demand could significantly change the binding constraints. In winter, PV generation is lower, meaning the DOE limits would be higher (less voltage rise pressure). Using a single week's DOE profile cycled annually therefore **overestimates curtailment** in winter and **may underestimate** it during peak summer weeks with atypically high irradiance. A more robust implementation would compute the DOE for multiple representative weeks, or run a full-year SOCP.

6.3 Uniform vs. Differentiated DOE

The SOCP-OPF produces **differentiated DOEs**: each node receives a different limit reflecting its network position and phase. This is economically superior to a uniform DOE (which would give every node the most restrictive limit in the network). Table 7 shows that five nodes (L10, L22, L24, L16, L27) are effectively unconstrained for residential-scale PV. Applying a uniform DOE of ~2.5 kW to all ten nodes would needlessly curtail these five nodes' export, reducing prosumer welfare without benefit to the network.

6.4 SOC Continuity Between Days

The original code reset the battery SOC to 50% at the start of every day, making each day's optimisation fully independent. This was corrected in the updated implementation so that the

end-of-day SOC is carried forward as the initial condition for the next day (with the terminal constraint applied only on the final day of the year). The impact is most significant around multi-day weather events: a string of cloudy days depletes the battery, and the corrected model allows the next sunny day's dispatch to reflect that depleted state rather than naïvely assuming a full 50% start.

6.5 Export Constraint and Arbitrage

The constraint $p_{export} \leq p_{pv}$ prevents battery-to-grid export. While this reflects a common network operator preference (to avoid batteries operating as uncontrolled inverters), it forecloses potentially profitable evening export during the peak tariff window (35 ¢/kWh vs. 8 ¢/kWh FiT). Relaxing this constraint in a future iteration — subject to DOE limits — could substantially increase prosumer revenue while remaining network-safe.

6.6 Phase Imbalance and DOE Design

The SOCP was solved independently per phase (three uncoupled problems), which is a simplification of the full three-phase BFM. In reality, mutual inductance between phases means that export on phase A affects voltages on phases B and C through the zero-sequence path. The single-phase approach underestimates cross-phase coupling and may therefore slightly overestimate DOE limits for phase A nodes. A three-phase BFM SOCP would yield more conservative and physically accurate results, at the cost of greater model complexity.

7. Conclusion

This study implemented a complete active distribution network management pipeline for a 500 kVA LV feeder with 36 prosumer nodes. The LP-based HEMS demonstrates that battery storage can smooth household net load, reduce annual feeder export and reverse power flow, and lower electricity cost relative to PV-only operation. However, the full-year OpenDSS results show that household-level optimisation alone does not guarantee improved voltage quality under all operating conditions. The SOCP-OPF Dynamic Operating Envelope framework provides network-safe, location-aware export limits that are superior to uniform caps: five of the ten HEMS nodes are entirely unconstrained for residential-scale PV, while Phase A nodes at the feeder end are limited to 1–3 kW due to cumulative voltage rise. The OpenDSS full-year verification confirms that applying the DOE reduces the maximum voltage to 1.0490 p.u. and

lowers annual feeder export to 9.313 MWh, demonstrating its effectiveness for managing PV-driven voltage rise and reverse power flow. However, two rare undervoltage violation steps remain, with the minimum voltage reaching 0.8855 p.u. This indicates that export envelopes mainly address overvoltage and reverse-power-flow issues, while additional voltage-support measures may be required to address low-voltage events under high demand conditions.

Github URL: https://github.sydney.edu.au/GregorVerbicGroup/ELEC5208_Task3_Group36

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